Combustion Enhancement by Axial Vortices

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A tapered slot jet was studied experimentally in nonreacting and reacting tests using hot-wire anemometry, water-tunnel flow visualization, and planar laser-induced fluorescence imaging. The tapered slot jet is a modified elliptic jet with a conical contraction leading to its elliptic outlet. The added contraction changed the entire flowfield relative to a regular elliptic jet. The jet spread in the major axis plane was larger than in the minor axis plane, which is the opposite behavior of an elliptic jet. The elliptic jet spreads much faster at the minor axis plane relative to the major axis plane, leading to axis switching downstream of the nozzle. In the tapered slot jet, no axis switching was observed. The turbulence amplification in the jet core was higher than in circular and elliptic jets. The different behavior was attributed to the change in flow direction, inside the nozzle, from the conical section to the slot outlet. During this transition, the flow acquired angular momentum, thereby generating axial vorticity. The contraction angle and the outlet aspect ratio were found to affect the spreading rate of the jet at both axis planes and to vary the turbulence augmentation. The effect of the increased turbulence on reactive flow was tested in a premixed flame. The combustion rate was augmented in both the core and edges of the flame relative to a circular burner.

Nomenclature

a = major axis of the nozzle
$Æ$ = aspect ratio
b = minor axis of the nozzle
D_e = equivalent diameter (circular nozzle area equal to the elliptic nozzle area)
L = nozzle length
R = radial coordinate
$R_{0.5}$ = half-width of the jet (radial distance where $\bar{U} = U_{\mathfrak{C}}/2$)
Re = Reynolds number
U = mean axial velocity
u' = rms of fluctuating axial velocity
x = axial coordinate
α = contraction angle inside the nozzle

Subscripts

o = conditions at the exit
$\mathfrak{C}$ = centerline
SL = shear layer

Introduction

THE combustion process requires adequate mixing between fuel and oxidizer and proper flameholding conditions to initiate and sustain the reaction between them. Efficient combustion can be achieved by optimizing the mixing and the reaction processes. When the fuel and oxidizer are initially separate, it is of special importance to produce sufficient large-scale mixing. After the reactants are brought together to the same volume on a gross scale, enhanced mixing on the molecular level is needed to accelerate the reaction rate. Both large-scale mixing and molecular mixing are governed by the flow dynamics. In reacting systems that contain mixing layers, large-scale mixing is governed by large-scale coherent structures. These structures dominate the mixing layer evolution in both low and high Reynolds number flows.[1-3] Molecular mixing is related to the fine-scale turbulence produced during the development of the mixing layer.

Much research has been done to develop different active and passive methods of augmenting large- and small-scale mixing. Recent reviews have summarized these efforts.[2-4] One of the passive methods investigated was the use of small-aspect-ratio elliptic nozzles.[5] The large-scale mixing of this jet was substantially increased, relative to a circular jet, as a result of self-induction of the elliptic vortices. The small-scale mixing was also enhanced using this jet, a result of high azimuthal instability modes amplified in this noncircular configuration. The small-aspect-ratio elliptic nozzle was also shown to have good mixing characteristics in high Reynolds number reacting flows.[6] A modified configuration of the elliptic nozzle was developed later to improve small-scale mixing in the circumference and core region of the jet. The modified nozzle, called the tapered slot nozzle, consists of a conical contraction region that leads into the elliptic outlet. The basic flow characteristics of the jet from this nozzle have been described by Gutmark and Schadow.[7] The tapered lost jet's growth pattern was very different from that of the elliptic jet. The tapered slot jet spreads more to the major axis side than to the minor axis side, which is the opposite growth characteristic to the elliptic jet.[5,8] Consequently, the axis switching typical of an elliptic jet was not observed in this jet. The growth rate of the turbulence level in the core region, as well as in the jet's circumference, was higher than in the circular and elliptic jets. The increased turbulence production was attributed to the axial vorticity component generated by the nozzle's design.

In the present study, the effect of the nozzle design parameters on the tapered slot jet's flowfield are reported in nonreacting flows, and the combustion characteristics of a tapered elliptic flame are visualized using planar laser-induced fluorescence (PLIF).

Experimental Setup

Three different experimental methods were used in this work to study the tapered slot jet.

1) Nonreacting freejet tests were performed to study the shear-flow dynamics with hot-wire anemometry.

2) Flow visualization tests were concluded in water-tunnel tests.

3) The combustion dynamics were visualized using PLIF imaging in a laboratory burner.

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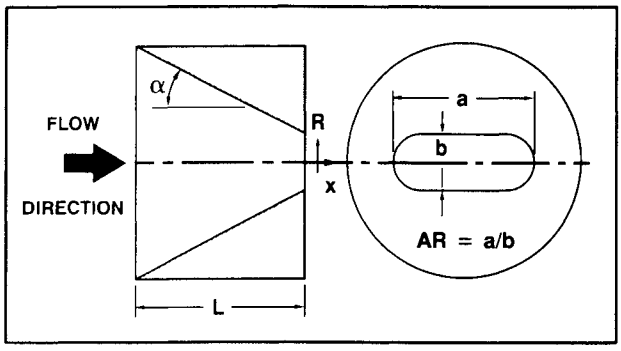

Fig. 1 Tapered elliptical nozzle design.

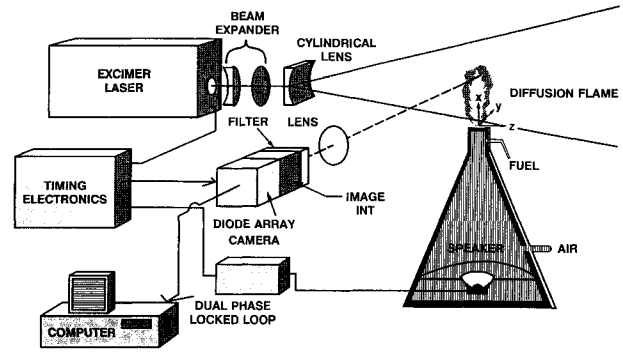

Fig. 2 Experimental setup for PLIF OH-imaging of a burner.

Table 1 Tapered slot nozzle geometrical configurations

Method	Nozzle no.	Geometrical parameters: Aspect ratio, a/b	Length, cm	Contraction angle α, deg	Equivalent diameter, cm
Air tests	1	3:1	3.67	30	3.95
	2	3:1	7.87	15	3.95
	3	3:1	16.13	7.5	3.95
	4	2:1	5.99	15	4.76
	5	5:1	9.53	15	3.28
Water test	6	3:1	8.20	23	4.80
Burner	7	3:1	2.54	22	1.96

The freejet tests were conducted using a tapered nozzle that had a conical contraction leading to the elongated outlet slot. The outlet slot replaced an elliptic shape to simplify the design and manufacturing. The slot consisted of a rectangular middle section with half-circles, the diameters of which were equal to the slot's width, attached to each end of the rectangle (Fig. 1). Five different tapered nozzles were tested in the air jet tests and compared to a circular nozzle. Their geometrical parameters are summarized in Table 1. The contraction angle α is in the radial direction only. The air was supplied by a blowdown facility. The jet exit velocity U_o was typically 50 m/s, corresponding to a Reynolds number of 1.6×10^5.

Hot-wire anemometers were used for probing the flowfield; the measurements included mean velocity contours and the distribution of turbulent intensities. A computer-controlled, three-axis traverse mechanism was used to scan the flowfield at 400 measurement points. The hot-wire calibration, data acquistion, and analysis were done on a VAX 750 minicomputer.

The flow visualization test were conducted in a water tunnel using dye injection and air bubbles and also nozzle 6 in Table 1. The flow was illuminated with a slot lamp. Flow excitation was achieved using a butterfly valve in the inlet duct. The exit velocity was 0.5 m/s, yielding $Re = 20{,}000$.

The flow characteristics of the propane/air burner were visualized by PLIF imaging of in situ OH radicals. In this technique, which is shown schematically in Fig. 2 and described in detail in Ref. 9, an excimer laser beam was expaned into a thin sheet to excite OH molecules in the premixed flame of the burner. The fluorescence was imaged with an intensified diode array camera. The burner can be acoustically excited with varying phase shifts with respect to the laser. Changing the phase angle can serve as a means of stroboscopically mapping out the time dependence of the flame structure. The burner (nozzle 7) had an aspect ratio of 3:1 and a contraction angle of 22 deg. The air velocity was 5 m/s, corresponding to a Reynolds number of 6500.

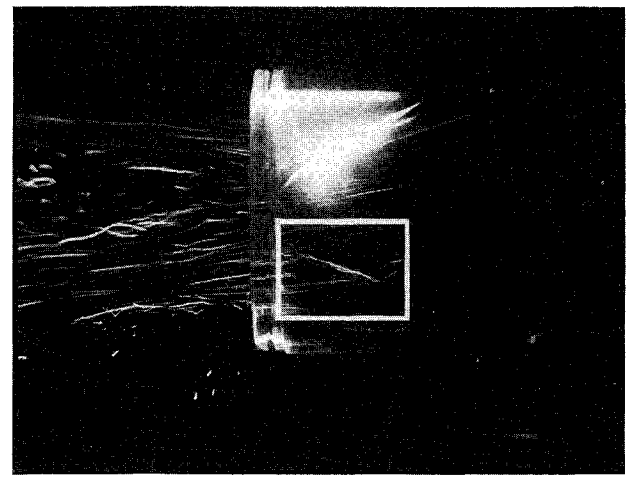

Fig. 3 Visualization of the flow pattern inside the tapered elliptic nozzle (water-tunnel test). The generation of axial vortices occurs at the intersection between the conical section and the slot surfaces (marked in the photograph).

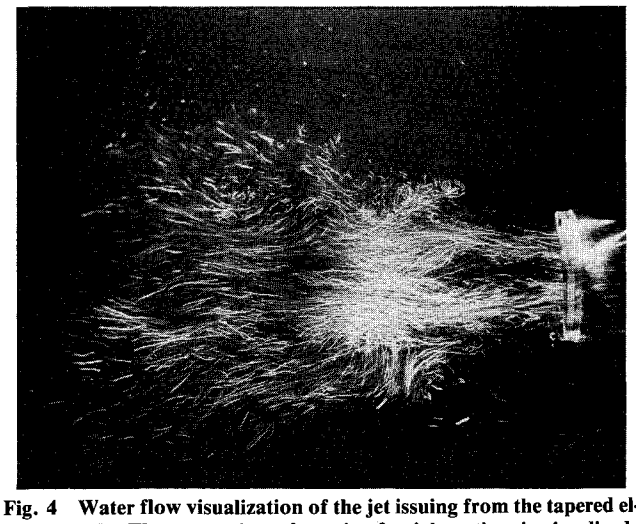

Fig. 4 Water flow visualization of the jet issuing from the tapered elliptic nozzle. The generation of a pair of axial vortices is visualized.

Results and Discussion

Flow Visualization

Water-tunnel tests were conducted to visualize the flow pattern of the tapered slot jet inside the nozzle and in the downstream jet flow. Previous air test results showed that the tapered slot jet has a higher turbulence intensity in the jet center relative to a circular jet.[7] The flow characteristics varied considerably around the jet's circumference. These features were attributed to the effect of the tapered nozzle on the

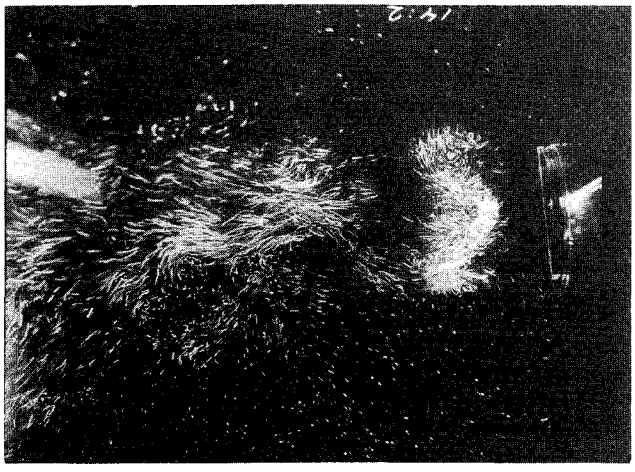

Fig. 5 Water flow visualization demonstrating the deformation of the initial vortices of the tapered elliptic jet.

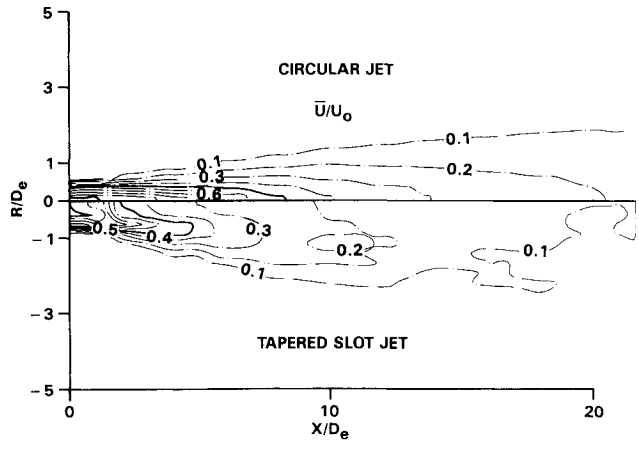

Fig. 6 Mean axial velocity contours.

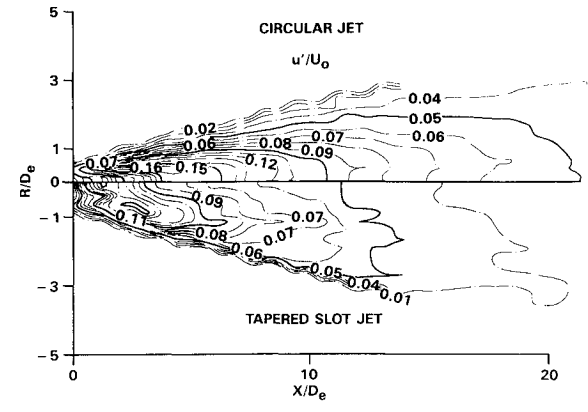

Fig. 7 Axial tubulence intensity contours.

outgoing flow. The flow streamlines inside the nozzle change from the conical contraction section to the slot outlet. During this transition, the flow acquires a vorticity component in the axial direction. On the intersection region inside the nozzle, between the conical contraction and the slot, axial vortices are formed as shown in Fig. 3. Two pairs of intense axial vortices are generated in the jet. One such pair, which is shown in Fig. 4, was visualized by illuminating the jet along its longer dimension and off its axis. The axial vortices deform the outgoing circumferential vortices (Fig. 5), which then obtain complex azimuthal structure. The jet evolves into secondary jets located at the two sides of the major axis.

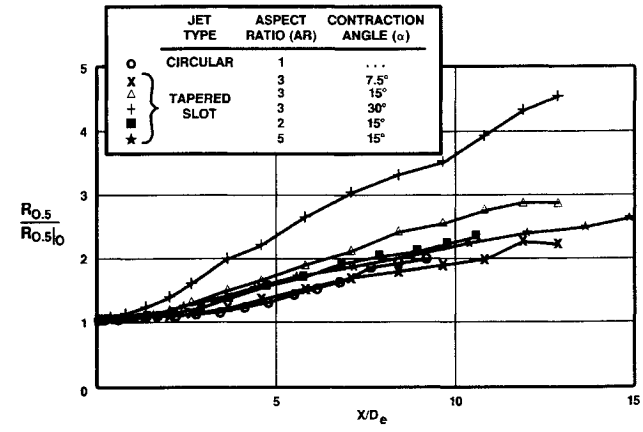

Fig. 8 Variation of jet spreading rate with contraction angle and aspect ratio (major axis).

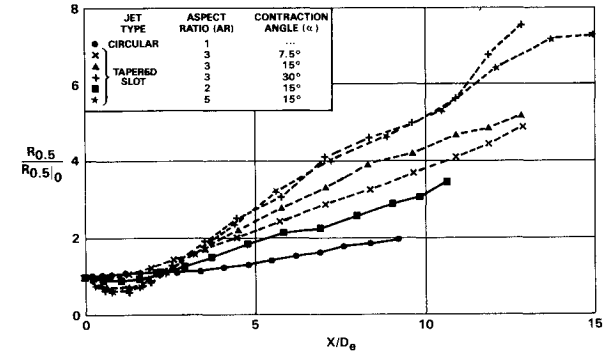

Fig. 9 Variation of jet spreading rate with contraction angle and aspect ratio (minor axis).

Flowfield

The mean and turbulent flowfields of the tapered slot jet are compared to a circular jet in Figs. 6 and 7. In the tapered slot, the centerline mean velocity decays faster relative to a circular jet. At the same time, higher mean velocity extensions are developed in the tapered slot jet's circumference, while the circular jet has a Gaussian-shaped mean velocity in the radial direction, decreasing monotonically to zero at the jet's edge. The axial turbulence intensity contours, given in Fig. 7, show the rapid amplification of the turbulence level along the centerline of the tapered slot jet relative to the low rate of the circular jet, such that a much larger turbulence intensity exists in this jet for $x/D_e \leq 4$. The circular jet centerline turbulence intensity becomes larger than that of the tapered slot jet for $x/D_e > 4$. Both jets have similar turbulence growth along their shear layers.

Two geometrical parameters of the nozzle design determine the jet characteristics: 1) the conical contraction angle α, and 2) the aspect ratio $\mathcal{R}$ of the outlet slot. In the following section, the effect of these parameters is discussed.

Effect of Conical Contraction Angle

The change of jet spreading rate in the major and minor axis planes for different contraction angles is shown in Figs. 8 and 9, respectively. The spreading rate in the major axis plane increases with increased contraction angle. The jet with the smallest contraction angle ($\alpha = 7.5$ deg) has the same spreading rate as a circular jet. The rate is doubled for $\alpha = 30$ deg. In the minor axis plane (Fig. 9), the three tapered slot jets spread faster than the circular jet. The same trend of increased spread with larger α is also obtained in this plane. The initial reduction of the jet width at the minor axis plane for $x/D_e \leq 2$ is due to the development of two secondary jets at the outer sections of the major axis.

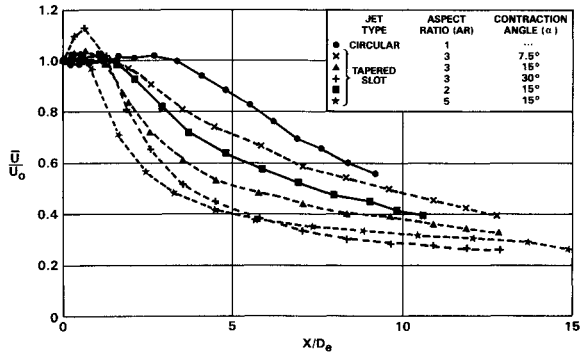

Fig. 10 Centerline velocity decay for different contraction angles and aspect ratios.

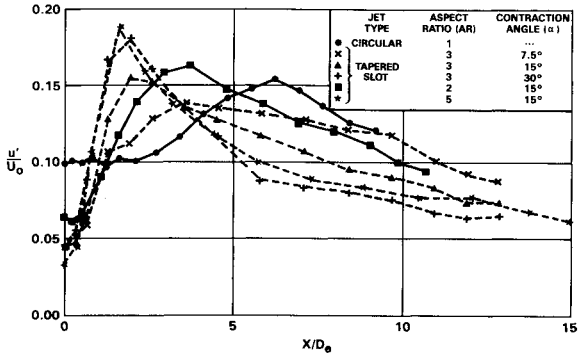

Fig. 11 Centerline turbulence intensity of different contraction angles and aspect ratios.

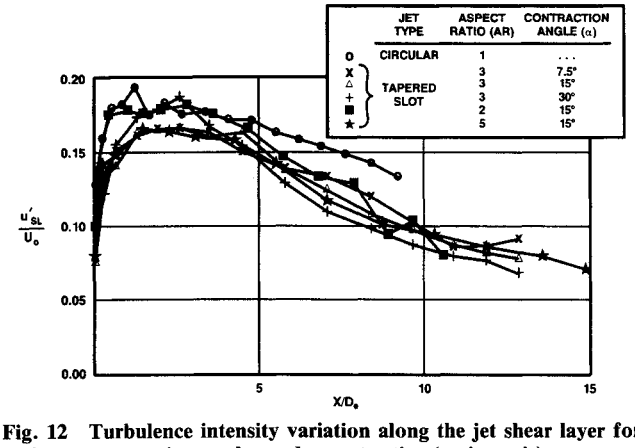

Fig. 12 Turbulence intensity variation along the jet shear layer for different contraction angles and aspect ratios (major axis).

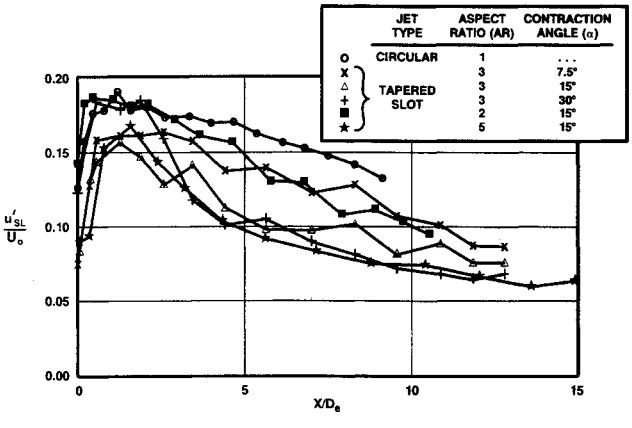

Fig. 13 Turbulence intensity variation along the jet shear layer for different contraction angles and aspect ratios (minor axis).

The variation of the spreading rate with the contraction angle affects the mean velocity decay along the jet centerline (Fig. 10). The highest decay rate correlates with the largest contraction angle, $\alpha = 30$ deg. The jet with the smallest angle, $\alpha = 7.5$ deg, is closest to the circular jet mean velocity change. The potential core length, as determined by the distance from the jet exit to the location where the velocity drops below the exit velocity, is shorter for all three tapered slot jets, $x = 1.3D_e$, relative to $3.5D_e$ for the circular jet. The increased velocity of the jets with $\alpha = 15$ and 7.5 deg in this section is a result of the initial reduction of the jet's width at the minor axis side.

One of the important features of the tapered slot jet is the higher augmentation of turbulence in the core region. The contraction angle affects this behavior also (Fig. 11). The maximum amplification is obtained for the steepest contraction angle, $\alpha = 30$ deg. The turbulence level is increased from an initial value of $u'/U_o < 0.05$ at the exit, to u'/U_o at $x/D_e = 1.9$. The circular jet retains its initial level of 10% in this range, and it reaches its highest level of 15% at $6.2D_e$. The jet with the intermediate angle, $\alpha = 15$ deg, reaches the same peak level as the circular jet at $1.9D_e$, and the jet with the smallest angle has a maximum turbulence level of less than that of the circular jet. Similar results were obtained for the other two components of the turbulent fluctuations, v' and w' (not shown here).

The turbulence intensity variation along the shear layer region at the major axis side (Fig. 12) is nearly equal for all three tapered slot jets and similar to the circular jet. The highest level, 18%, is obtained at $2.6D_e$ for the tapered jets and at $1.5D_e$ for the circular jet. At the minor axis shear layer (Fig. 13), the contraction angle changes the amplification rate. The jet with the largest angle, $\alpha = 30$ deg, has the same behavior as the circular jet, while the other two reach a level 15% lower than the circular jet.

Effect of Aspect Ratio

Three tapered slot jets with aspect ratios of 2, 3, and 5 were tested and compared to a circular jet. All jets had a contraction angle of 15 deg. This jet spread at the major axis plane did not change with the aspect ratio, and it is slightly higher than the circular jet. The spread in the minor axis plane (Fig. 9) increases with the jet aspect ratio. This result is similar to rectangular jets.[10] The high-aspect-ratio jet ($\mathcal{R} = 5$) spreads almost twice as much as the low-aspect-ratio jet ($\mathcal{R} = 2$). All tapered jets grow in the minor axis plane more than the circular jet.

The high-aspect-ratio jet also has a higher mean velocity decay rate along the centerline (Fig. 10). The potential core length is reduced from $1.6D_e$ for the $\mathcal{R} = 2$ jet to $0.6D_e$ for the $\mathcal{R} = 5$ jet.

The highest amplification of turbulence intensity along the centerline is for the $\mathcal{R} = 5$ jet (Fig. 11). Its rate is even higher than that of the jet with the steepest contraction angle ($\alpha = 30$ deg, $\mathcal{R} = 3$). The jet reaches a turbulence level of nearly 19% at a distance of $1.6D_e$ from the nozzle. The amplification rate is reduced with decreasing aspect ratio but is still higher than the circular jet, even for the $\mathcal{R} = 2$ jet.

The amplification of the turbulent fluctuations in the shear layer are given in Figs. 12 and 13 for the major and minor axes of the tapered jet with different aspect ratios. The tapered jet with $\mathcal{R} = 2$ has the highest amplification rate at the major axis shear layer, which is also equal to that of the circular jet. The jets with aspect ratios of 2 and 3 have similar behavior and reach a maximum level that is lower than the circular jet. The same features are also observed at the minor axis plane (Fig. 13).

Reacting Flow

The tapered slot jet was tested in a premixed flame. Nozzle 7, with an aspect ratio of 3 and a contraction angle of $\alpha = 22$ deg, was used for the burner. Figure 14 shows the hydroxyl concentration in three axial cross sections of the flame.

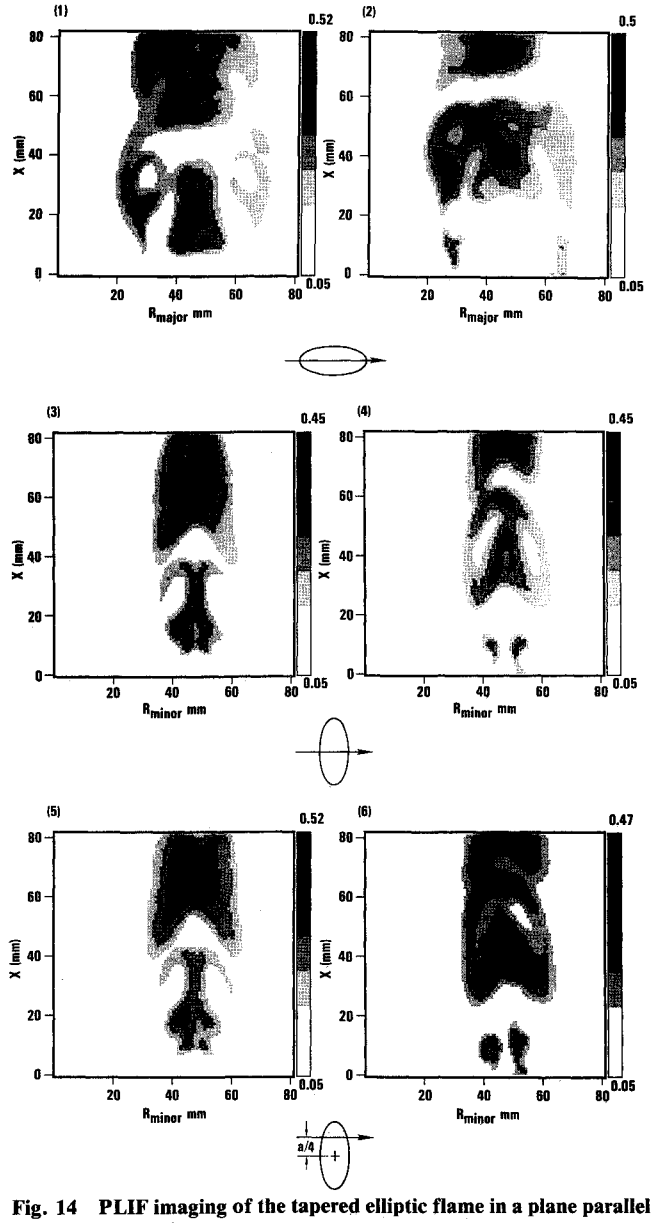

Fig. 14 PLIF imaging of the tapered elliptic flame in a plane parallel to the flame's axis.

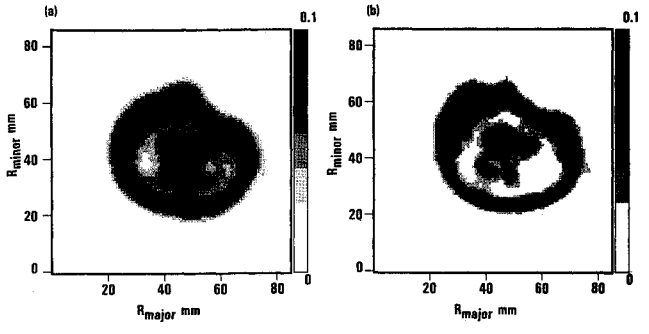

Fig. 15 PLIF imaging of the tapered elliptic flame in a plane perpendicular to the flame's axis.

Frames 1 and 2 are obtained by passing the laser sheet along the major axis in the flame's axis direction. In frames 3 and 4, the beam passes through the minor axis plane and, in frames 5 and 6, the beam plane is parallel to the minor axis plane at a distance equal to one-fourth the major axis length from the center. Frames 1, 3 and 5 differ from frames 2, 4, and 6 by the time delay between the laser pulse and the forcing signal; thus, the flow is visualized in the two groups at different phases relative to the forcing. The OH concentration is higher in this flame relative to a circular burner. The reaction occurs both in the flame circumference and in the flame core. This differs from a circular premixed flame in which the combustion is concentrated in the outer edge of the air jet.[9] The reaction in the flame's core is sustained as a result of improved small-scale mixing obtained by the axial vortices that are generated in this region. The structures in the jet's core are surrounded by shear layer vortices, which can be seen in frames 1, 4, and 6.

The special character of the tapered slot jet core is visualized in a different cross-sectional plane in Fig. 15. The laser sheet is "slicing" the flame perpendicular to the flame's axis. Figure 15a shows an averaged picture of 100 single realizations, and Fig. 15b shows an instantaneous picture of 18-ns duration. The three-dimensional structures in the middle of the flame are cross sections in the axial vortices emanating from the nozzle. The outer closed surface is a cross section of the flame's edge. These pictures show the increased OH concentration obtained by the additional turbulence generation gained by using this burner.

Summary and Conclusions

A novel design of an injector nozzle aimed at reducing flow coherence and enhancing turbulence and small-scale mixing in the entire reactor is described in this paper. An elliptical jet, which was shown to have special mixing characterisitics relative to a circular jet, was modified by adding a conical contraction leading to the elongated slot outlet. The combination of this contraction with the noncircular outlet has a substantial effect on the outgoing jet flow. The elliptical jet[5] is characterized by the switching of its axes in the downstream direction. The initial elliptic jet changes shape very rapidly in response to the high spreading rate at the minor axis section. The initial minor axis becomes the jet's new major axis, and vice versa at the other side. The flow of the tapered slot jet is very different. The spreading rate on the side of the minor axis is reduced, while the growth along the major axis is increased considerably. At the same time, the flow accelerates at the major axis plane, resulting in the formation of a dobule-jet flow.

The centerline turbulence level of the tapered slot jet is up to five times higher than in the circular jet. In addition to the highly turbulent flow in the vicinity of the jet center, the high turbulence activity in the jet's shear layer, i.e., at the jet periphery, is maintained.

These flow characteristics are related to several different causes. The introduction of the nonsymmetric azimuthal disturbances into the initial shear layer of the jet results in a very complicated flowfield. The flow is unstable to numerous azimuthal modes, leading to an eventual breakdown of the flow into small-scale turbulence, thus enhancing the mixing at the fine scales. The double shear layer generated inside the flow by the two jets that are formed in the major axis plane contributes to the additional production of turbulence in the shear layer, as well as in the core region. The third cause, probably the dominant one, is related to the inside structure of the nozzle. The intersection between the conical contraction and the slot introduces axial rotation into the outcoming flow. The jet flow injected from the nozzle has, in addition to the spanwise vorticity, an axial component. The angular momentum of the axial vortices is determined by the transition surface between the conical contraction and the slot outlet. It is shown that by increasing either the conical contraction angle or the slot aspect ratio, the intensity of the axial vortices is increased and the ensuing effect on the mean flowfield is higher. These two geometrical changes have similar effects on the above-mentioned transition surface. The two parameters can be used to control and vary the mean and turbulent flow of the tapered slot jet as desired. The spreading rate of the jet at the two axis planes can be altered, as well as the turbulence level in the jet's core and shear layers.

The structure of the tapered slot burner flame is influenced by this special flowfield. Intense combustion is observed in both the shear layer and core regions, as is expected from the cold flow results. Because of the enhanced small-scale mixing of this jet, the flame burns the fuel efficiently without soot formation, even at high equivalence ratios.

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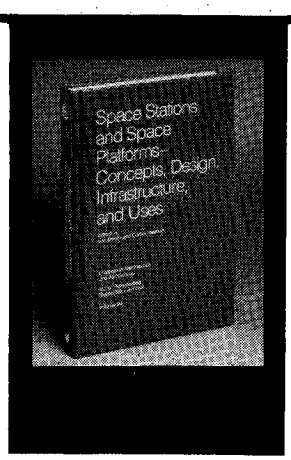

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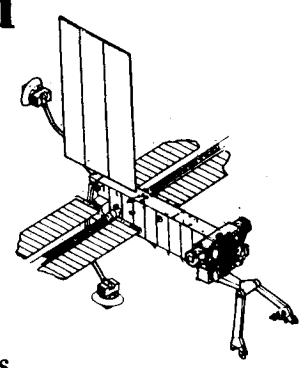

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